INTERNAL PROJECT PROPOSAL

HYBRID MARKET INTELLIGENCE SYSTEM (MIS)

To: CTO, Kerchanshe Group
From: Marketing Intelligence Officer
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1. Executive Summary

To maintain and strengthen Kerchanshe Group's market leadership position, the organization requires a centralized, real-time, data-driven Market Intelligence System (MIS) capable of capturing, processing, analyzing, and reporting market information from the field.

This proposal outlines the development of an internally owned and customized enterprise-grade Market Intelligence System built on modern open-source technologies and fully integrated with Microsoft Dynamics 365 Finance & Operations (D365FO).

The system will provide:

- Real-time field market intelligence
- Competitor pricing monitoring
- Distributor and customer feedback tracking
- Automated survey management
- Multimedia evidence collection
- Advanced analytics and reporting
- Power BI dashboards
- Direct integration with D365FO master and transactional data

By developing the platform internally, Kerchanshe Group gains full ownership of business intelligence assets while significantly reducing recurring software subscription costs.

2. Strategic Objectives

The primary objectives of the proposed system are:

2.1 Enhance Market Visibility

Provide real-time visibility into:

- Competitor activities
- Product availability
- Distribution performance
- Retail execution
- Customer sentiment

2.2 Improve Decision-Making

Transform field observations into actionable business intelligence that enables:

- Faster commercial decisions
- Improved pricing strategies
- Distribution optimization
- Market expansion planning

2.3 Eliminate Vendor Dependency

Build on open-source frameworks to:

- Avoid expensive license costs
- Eliminate vendor lock-in
- Maintain complete system flexibility

2.4 Ensure Data Sovereignty

Maintain complete ownership and control of:

- Customer insights
- Competitor intelligence
- Market research data
- Survey responses

2.5 Integrate Field Intelligence with ERP

Bridge operational market data with financial and commercial data available in D365FO.

3. Business Benefits

Operational Benefits

- ✔ Faster market response
- ✔ Better visibility into competitor actions
- ✔ Improved product placement monitoring
- ✔ Reduced manual reporting
- ✔ Standardized field data collection
- ✔ Automated alerting mechanisms

Financial Benefits

- ✔ Reduced software licensing costs
- ✔ Improved sales performance
- ✔ Better pricing decisions
- ✔ Reduced market intelligence outsourcing expenses
- ✔ Improved revenue protection

Strategic Benefits

- ✔ Competitive advantage
- ✔ Data-driven culture
- ✔ Unified intelligence platform
- ✔ Stronger market leadership

✔ Scalable digital foundation

4. Proposed System Architecture

The solution consists of five major layers:

Layer 1: Mobile Intelligence Platform

Field agents use mobile devices to collect:

Data Types

- Outlet surveys
- Competitor pricing
- Product availability
- Distributor feedback
- Customer interviews
- Market observations

Mobile Features

- Offline-first architecture
- GPS tracking
- Photo capture
- Video recording
- Voice recording
- Digital forms
- Auto synchronization

Layer 2: Administrative Web Portal

Provides management and administration capabilities.

Features

Survey Builder

Create dynamic surveys without coding.

User Management

Manage:

- Users
- Roles
- Permissions
- Territories

Data Validation

Monitor:

- Missing records
- Duplicate entries
- Invalid submissions

Report Builder

Drag-and-drop report generation.

Layer 3: Central Application Server

Acts as the intelligence processing engine.

Core Functions

Data Validation Engine

- Duplicate detection
- Data completeness checks
- Business rule validation

Workflow Engine

Automates:

- Approvals
- Notifications

- Escalations

Analytics Engine

Processes:

- Survey responses
- Competitor pricing
- Trend calculations

Sentiment Analysis Engine

Processes interview recordings to:

- Convert speech to text
- Identify sentiment
- Extract themes

Layer 4: Central SQL Database

Stores:

Master Data

- Customers
- Distributors
- Products
- Locations

Transaction Data

- Surveys
- Interviews
- Observations
- Media files

Audit Data

- User activity logs

- Synchronization history
- Security records

Layer 5: Analytics & Reporting Layer

Powered by Power BI.

Dashboard Categories

Executive Dashboard

- Revenue trends
- Market share indicators
- Distribution coverage

Sales Dashboard

- Territory performance
- Outlet penetration
- Product movement

Competitor Dashboard

- Pricing comparison
- Availability analysis
- Promotion tracking

Customer Insights Dashboard

- Sentiment analysis
- Complaint trends
- Customer satisfaction

5. D365FO Integration Framework

The proposed MIS will integrate directly with Microsoft Dynamics 365 Finance & Operations.

Master Data Synchronization

The system will synchronize:

D365FO Entity	MIS Usage
Customers	Survey Mapping
Vendors	Supplier Analysis
Products	Product Intelligence
Sales Territories	Geographic Analysis
Workers	User Assignment

Transactional Integration

The MIS will consume:

- Sales transactions
- Inventory levels
- Customer performance
- Distribution records

This allows field observations to be validated against actual business performance.

Key Integration Scenarios

Competitor Pricing Alert

Field agents report competitor prices.

System compares:

Competitor Price vs Kerchanshe Price

If threshold exceeded:

- Alert generated
- Management notified

Product Availability Monitoring

Field reports indicate:

- Out of stock
- Poor visibility
- Product displacement

System validates against sales and inventory data from D365FO.

Distributor Performance Monitoring

Field observations combined with:

- Revenue
- Order frequency
- Product availability

to produce distributor performance scores.

6. Mobile Application Design

Home Dashboard

Displays:

- Assigned tasks
- Daily targets
- Recent activities
- Synchronization status

Survey Module

Supports:

- Dynamic questionnaires
- Mandatory fields
- GPS validation
- Offline storage

Interview Module

Supports:

- Audio recording
- Voice notes
- Automatic transcription
- Sentiment extraction

Multimedia Module

Capture:

- Photos
- Videos
- Shelf displays
- Competitor products

with automatic metadata tagging.

Synchronization Module

Features:

- Offline queue
- Background synchronization
- Conflict resolution
- Retry mechanisms

7. Administrative Portal Features

Dashboard Center

Provides:

- Real-time KPIs
- Intelligence summaries
- Operational alerts

Survey Management

Administrators can:

- Create surveys
- Schedule surveys
- Publish surveys
- Archive surveys

User Administration

Manage:

- Users
- Roles
- Teams
- Territories

Report Builder

Create reports using:

- Product filters
- Geographic filters
- Time filters
- Customer filters

Export formats:

- PDF
- Excel
- PowerPoint

8. Smart Processing & Automation

Automated Data Cleaning

Automatically:

- Remove duplicates
- Detect anomalies
- Validate completeness

Multimedia Analysis

Future enhancement:

AI-based image recognition to identify:

- Shelf visibility
- Product placement
- Competitor presence

NLP & Sentiment Analysis

Recorded interviews processed through:

- Speech-to-text
- Translation
- Sentiment scoring
- Theme extraction

Automated Alerts

Examples:

Pricing Gap Alert

Competitor price lower than threshold.

Stockout Alert

Multiple field reports indicate shortages.

Product Displacement Alert

Competitor replacing Kerchanshe products.

Distribution Risk Alert

Declining distributor engagement.

9. Security & Governance

Security Controls

- Role-based access
- Multi-factor authentication
- Encrypted communication
- Secure API gateway

Audit Controls

Track:

- User activity
- Data modifications
- Synchronization events

Compliance

Ensure compliance with:

- Corporate governance policies
- Data protection standards
- Internal audit requirements

10. Technology Stack

Layer	Technology
Mobile App	Flutter
Web Portal	React.js
Backend API	.NET Core
Database	SQL Server
Reporting	Power BI

Integration	OData / REST APIs
Authentication	Azure AD
ERP Integration	D365FO

11. Estimated Project Phases

Phase	Duration
Requirements Gathering	1 Weeks
System Design	1 Weeks
Development	2 Weeks
Integration	2 Weeks
UAT	1Weeks
Go-Live	1 Weeks

Total Estimated Duration: 8 Weeks

12. Expected Outcomes

Upon implementation, Kerchanshe Group will achieve:

- Single source of market intelligence
- Faster commercial decisions
- Improved competitive visibility
- Better pricing governance
- Stronger distributor management
- Reduced operational costs
- Enhanced revenue protection
- Full integration with D365FO

13. Recommendation

The proposed Hybrid Market Intelligence System (MIS) provides Kerchanshe Group with a scalable, cost-effective, and strategically valuable platform that transforms field observations into actionable business intelligence.

The solution aligns with the Group's digital transformation strategy and creates a direct bridge between market realities and enterprise decision-making.

Approval is hereby requested to proceed with the System Requirements Document (SRD) phase and initiate detailed solution design and development planning.

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